Exercise Sheet 19

Use $(g\cdot\ell)(X)=\ell(\mathrm{Ad}(g^{-1})X)$ and $\omega_\ell(\xi_X,\xi_Y)=\ell([X,Y])$. Heisenberg coordinates are matrix coordinates, with law

$$(a,b,c)(a',b',c')=(a+a',b+b',c+c'+ab').$$

Exercises 19.1–19.5 are core. Exercise 19.6 is an optional further example.

Exercise 19.1 (An affine comparison). The connected affine group has matrices $g(a,b)=\left(\begin{smallmatrix}a & b\\ 0 & 1\end{smallmatrix}\right)$, $a>0$. Its algebra has $[X,Y]=Y$ and

$$\mathrm{Ad}(g(a,b))X=X-bY,\qquad \mathrm{Ad}(g(a,b))Y=aY.$$

(a) With $p=\ell(X)$ and $q=\ell(Y)$, compute $g(a,b)\cdot\ell$ and list its orbits.

(b) On either two-dimensional orbit, find ξ_X,ξ_Y and the symplectic form in coordinates p,q.

(c) Compute the lower central series. Does the nilpotent orbit theorem apply?

Exercise 19.2 (Pullback of the orbit form). Fix $\lambda\neq0$ and $\ell_0=(p_0,q_0,\lambda)$ for the Heisenberg group. Let $Q:H\to\mathcal{O}_{\ell_0}$ be the orbit map.

(a) Show that $Q^*\omega_\lambda=\lambda\,da\wedge db$.

(b) Verify that $\theta=dc-a\,db$ is left invariant and that $Q^*\omega_\lambda=-\lambda\,d\theta$.

(c) Find a primitive for ω_λ on the plane, that is, a 1-form whose exterior derivative is ω_λ. Does the pullback identity imply that the orbit is three-dimensional?

Exercise 19.3 (A circle centre). Let $\Gamma=\{(0,0,2\pi k):k\in\mathbb{Z}\}$ and $\bar H=H/\Gamma$.

(a) For which nonzero λ does the Schrödinger representation

$$\Pi_\lambda(a,b,c)u(t)=e^{i\lambda(c+bt)}u(t+a)$$

descend to $\bar H$?

(b) Which point-orbit characters descend? Why is it incorrect to attach an irreducible representation of $\bar H$ to every Heisenberg plane without a further condition?

Exercise 19.4 (Moving within a plane). Let $\ell=(p,q,\lambda)$ with $\lambda\neq0$ and use $\mathfrak{m}=\mathrm{span}\{Y,Z\}$.

(a) Compute χ_ℓ on M and the induced action in the section $s(t)=(t,0,0)$.

(b) Set $(T_q\varphi)(t)=\varphi(t-q/\lambda)$. Show directly that $T_q\Psi_{0,\lambda}(g)=\Psi_{q,\lambda}(g)T_q$.

(c) Explain why p does not enter this induced formula and why different (p,q) do not give different irreducibles at fixed nonzero λ.

Exercise 19.5 (Hamiltonian coordinates). On a Heisenberg plane, define the Hamiltonian vector field X_f of a smooth real function f by $\iota_{X_f}\omega_\lambda = df$. This means $\omega_\lambda(X_f, v) = df(v)$ for all tangent vectors v. Define $\{f, h\} = \omega_\lambda(X_f, X_h)$.

(a) Compute X_f and $\{f, h\}$ in coordinates.

(b) Check $\{p, q\} = \lambda$. For $\mu_W(\eta) = \eta(W)$, verify $\{\mu_X, \mu_Y\} = \mu_Z$ and the two central brackets.

Exercise 19.6 (Optional extension: a four-dimensional nilpotent algebra). Let $\mathfrak{g}$ have basis X, Y, Z, W with $[X, Y] = Z$, $[X, Z] = W$, and other basis brackets zero. Let G be its connected simply connected group. Write dual coordinates (p, q, z, r), and restrict attention to $r \neq 0$.

(a) Compute the infinitesimal coadjoint vectors and show that r and $C = 2rq - z^2$ are invariant.

(b) Show that the orbit through a functional with $r \neq 0$ is the surface on which r and C take that functional's values. Show also that (p, z) are coordinates on this surface, with $q = (C + z^2)/(2r)$.

(c) Find its symplectic form and a polarisation.